OpenCart E-Commerce Application

Database Testing Test Plan

Project: OpenCart Project

Document Version: 1.0

Testing Type: Database Validation

Approach: Manual Database Testing using SQL

This document defines the scope, objectives, strategy, environment, test areas, entry/exit criteria, risks, responsibilities, and deliverables for database testing of the OpenCart project.

1. Document Control

Field	Details
Project	OpenCart E-Commerce Application
Document	Database Testing Test Plan
Version	1.0
Testing Type	Database Testing
Test Approach	Manual testing with SQL validation
Application	Web-based e-commerce application
Status	Portfolio Project

2. Introduction

The purpose of this Database Testing Test Plan is to define how database-level validation will be performed for the OpenCart e-commerce application. Testing will verify that data entered or generated through the application is stored, updated, retrieved, and maintained correctly.

The database testing activities are designed as part of a QA portfolio project. Where database access is available, SQL queries will be used to validate application data against the expected database state.

3. Objectives

• Verify application data is stored correctly in the database.

• Verify updated application data is reflected in the relevant records.

• Validate accuracy and consistency between related tables.

• Identify duplicate, incorrect, missing, or inconsistent data where applicable.

• Validate relationships between customer, product, cart, order, and other business data.

• Use SQL queries to support backend validation of functional test results.

4. Scope

Area	Validation Focus
Customer	Registration, customer details, email and telephone information
Account Information	Updated customer profile information
Address Book	Customer addresses and updates
Products	Name, price, special price, status, stock and category relationships
Shopping Cart	Cart-related product, quantity and customer/session data where applicable
Orders	Order creation, customer reference, totals, status and dates
Order Details	Products, quantities, prices and related order information
Returns	Return-related records where applicable
Reviews	Product review data and customer/product relationships
Currencies	Currency configuration and related values where applicable
Data Integrity	Consistency, relationships, duplicates and orphaned records

5. Out of Scope

• Database server administration and infrastructure management.

• Database performance/load testing unless separately planned.

• Database penetration/security testing.

• Production database modification or deletion.

• Backup and disaster-recovery testing.

• Features not accessible or not configured in the selected test environment.

6. Test Strategy

Activity	Approach
Data Validation	Compare application data with corresponding database records.
Insert Validation	Verify valid application transactions create expected records.
Update Validation	Verify application changes update the correct fields.
Delete Validation	Verify deletion behavior according to requirements.
Integrity Validation	Check relationships and consistency between related records.
Negative Validation	Check invalid, blank, duplicate or unexpected input where applicable.
SQL Validation	Use SELECT, WHERE, JOIN, GROUP BY and aggregate functions as appropriate.

7. Test Types

• Data Integrity Testing

• Data Accuracy Testing

• CRUD Validation (Create, Read, Update, Delete)

• Referential Integrity Testing

• Data Consistency Testing

• Null / Mandatory Field Validation

• Duplicate Data Validation

• Backend Validation after UI Transactions

• Regression Database Validation

8. Database Areas / Entities

• Customer

• Customer Address

• Product

• Category

• Product-Category Relationship

• Shopping Cart

• Order

• Order Product / Order Details

• Order History / Status

• Return

• Review

9.SQL Skills Covered

SQL Area	Examples
Basic Retrieval	SELECT
Filtering	WHERE, AND, OR, IN, BETWEEN, LIKE
Sorting	ORDER BY
Grouping	GROUP BY, HAVING
Uniqueness	DISTINCT
Aggregate Functions	COUNT, SUM, AVG, MIN, MAX
Joins	INNER JOIN, LEFT JOIN
Subqueries	Nested SELECT statements
Data Modification	INSERT, UPDATE, DELETE — controlled test DB only

9. Test Environment

Component	Planned Environment
Application	OpenCart e-commerce application
Environment	Local / authorized test environment with database access
Database	Database used by the selected OpenCart test environment
Method	Manual UI actions + SQL queries
SQL Tool	Approved SQL client / database console
Test Data	Controlled QA test data
Browsers	Supported browsers used for application validation

10. Entry Criteria

• Testable OpenCart environment is available.

• Required database access/read permissions are available for QA.

• Database schema is available or can be inspected.

• Required test data and test accounts are available.

• Relevant manual functional scenarios are identified.

• SQL client/tool connectivity is verified.

11. Exit Criteria

• Planned DB test cases are executed.

• Critical database validation failures are investigated and documented.

• Defects are logged with evidence where applicable.

• Fixed defects are retested.

• Required regression validation is completed.

• Test results and observations are documented.

12. Roles and Responsibilities

Role	Responsibility
QA Tester	Prepare DB cases, write SQL validation queries, execute tests, compare UI/DB results.
Developer	Investigate application/database defects, provide fixes and technical clarification.

13. Defect Management

Database-related defects will be reported in Jira using the project defect workflow. A defect should include the related story/test case, environment, reproduction steps, expected result, actual result, SQL evidence where appropriate, severity, priority and status.

Suggested flow: Open → In Progress → Fixed → Ready for Retest → Retest → Closed. If the issue still exists, return it for fixing.

14. Risks and Mitigation

Risk	Impact	Mitigation
No DB access	Backend validation cannot be per	oUrsmeeadn authorized local/test DB or obtain read-only QA access.
Schema differs by versio	Queries may fail	Verify actual schema before final SQL queries.
Incorrect test data	False results	Maintain controlled documented test data.
Direct data modification	Data corruption	Prefer read-only queries; never modify production data.
Environment instability	Execution delays	Record environment issues and coordinate with the team.

15. Deliverables

• DB_Test_Plan.pdf

• DB_Test_Cases.xlsx

• SQL_Queries.sql

• DB_Test_Data.xlsx

• DB_Testing_Documentation.pdf

16. Project Limitation

This is a QA portfolio project. Database testing should be demonstrated only against an authorized local/test OpenCart database. The public OpenCart demo site should not be treated as providing direct database access. Actual database validation requires a controlled environment with authorized access.

17. Approval

Role	Name	Status	Date
QA Tester	Project Owner	Planned	_____
QA Lead	_____	Planned	_____
Project/Team	_____	Planned	_____